Support Vector Machine (SVM)

Goal: Classification.

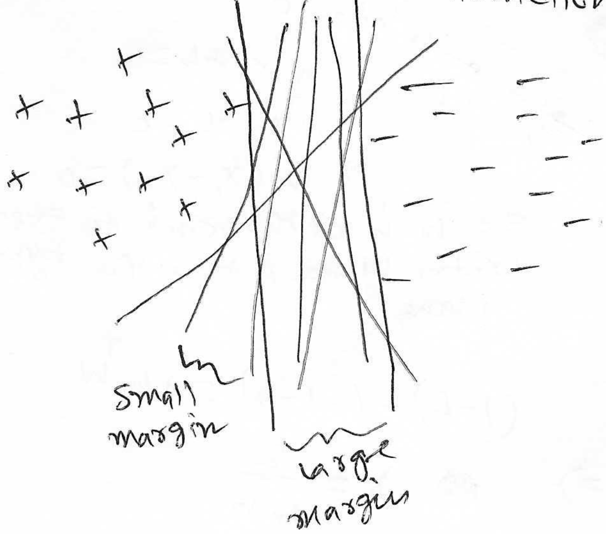
i) Generative approach ; (Naive Bayes) estimate $P(X|Y)$ & $P(Y)$ $\Rightarrow$ Bayes rule $\Rightarrow P(Y|X) \Rightarrow \hat{y}_{MAP}$

ii) Discriminative approach #1 : (Logistic regression) estimate $P(Y|X)$ $\Rightarrow \hat{y}_{MAP} \doteq \arg\max P(Y|X)$

iii) Discriminative approach #2 : (SVM) estimate score $(x) = w^T x$

$\hat{y} = \text{sign}(w^T x + b)$

Learn the discriminative function.



Many correct lines give 100% training accuracy.

which one to choose?

The one with the largest margin

reasons

- generalization bound
- other theoretical interpretations.

Notation: $y \in \{-1, +1\}$

$\vec{w}$ no longer has bias weight included.

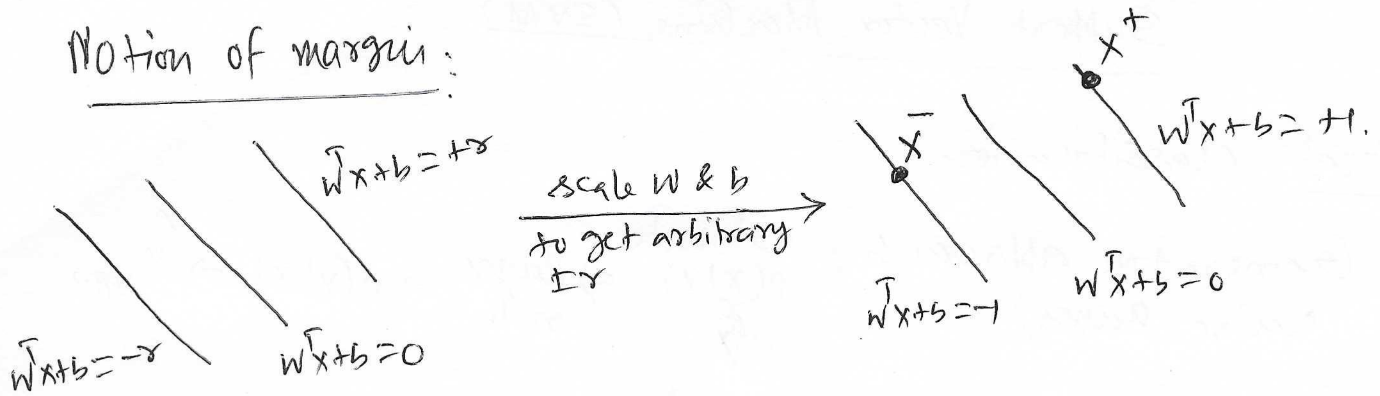
so $w^T x + b \geq 0$ rather than $w^T x \geq 0$

Goal: $\max_{\vec{w}, b}$ Margin , such that correct classification on training data.

what is correct classification?

$$\left.\begin{aligned} w^T x_i + b \geq 0 \quad \forall y = +1 \\ w^T x_i + b < 0 \quad \forall y = -1 \end{aligned}\right\} \Rightarrow y_i(w^T x_i + b) \geq 0$$

Notion of margin:



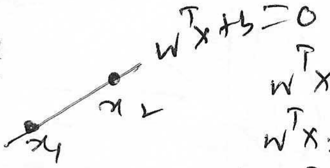
$$\left.\begin{array}{l} x^+ \Rightarrow \text{a point on } w^Tx+b=+1 \\ x^- \Rightarrow \quad " \quad " \quad " \quad w^Tx+b=-1 \end{array}\right\} \text{①}$$

Let a line from x^+ to x^- be orthogonal to our hyperplane $w^Tx+b=0$.

$\|x^+ - x^-\| \Rightarrow$ margin

$(x^+ - x^-) \perp$ hyperplane

$(x^+ - x^-) \parallel \vec{w}$ $\longrightarrow$ why? $\longrightarrow$



$w^Tx_1+b=0$

$w^Tx_2+b=0$

$\Rightarrow w^T(x_1-x_2)=0$.

so $\vec{w}$ is orthogonal to every vector lying within the hyperplane.

$$\Rightarrow (x^+ - x^-) = \underset{\uparrow \text{scalar}}{\lambda} \vec{w}$$

Multiplying by $\vec{w}$ on both sides,

$$w^T(x^+ - x^-) = \lambda w^Tw$$

$$\Rightarrow \underbrace{w^Tx^+}_{(1-b)} - \underbrace{w^Tx^-}_{(-1-b)} = \lambda w^Tw \quad \text{from eq}^n \text{ ①}$$

$$\Rightarrow \quad (1-b)-(-1-b) = \lambda w^Tw$$

$$\lambda = \frac{2}{w^Tw}$$

Hence, $\text{Margin} = \|x^+ - x^-\| = \dfrac{2}{w^Tw}\|\vec{w}\| = \boxed{\dfrac{2}{\|\vec{w}\|}}$ SVM with a hard margin

Fact: $\displaystyle\max_{\vec{w},b} \frac{2}{\|\vec{w}\|}$ is not easy to optimize with the constraint $y_i(w^Tx_i+b) \geq 0$

$$\Downarrow$$

$$\min_{\vec{w},b} \frac{1}{2}\|w\|^2 = \frac{1}{2}w^Tw$$

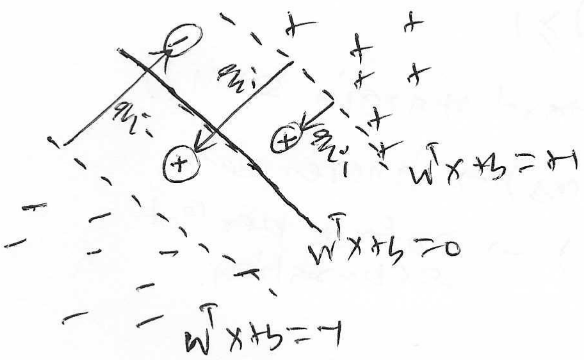
such that $y_i(w^Tx_i+b) \geq +1$ (why +1 instead of 0?)

This is solvable by quadratic programming.

Soft margin linear SVM.

What if the data is not linearly solvable?

Idea: Allow some "slack".



Allow every data point to violate constraint by some "slack" ξ_i.

$$y_i(w^Tx_i+b) \geq 1-\xi_i$$ (softening the constraint)

But we must penalize the violations. So we change the objective functions as,

$$\min_{\vec{w},b} \frac{1}{2}w^Tw + \underbrace{C}_{\text{some constant}}\underbrace{\sum_{i=1}^{N}\xi_i}_{\text{penalize the violations}} \text{ such that } y_i(w^Tx_i+b) \geq 1-\xi_i \ \forall i$$

$$\xi_i \geq 0 \quad (\text{why?})$$

C controls trade-off between:

- model simplicity (large margin) $\rightarrow \frac{1}{2}\|w\|^2$
- Constraints violation (errors) $\rightarrow \sum \xi_i$

Case-I $\quad C=0. \Rightarrow \min_{\vec{w},b,\xi} \frac{1}{2}\|w\|^2 \rightarrow \xi_i$ are free.

optimization: minimize $\|w\|^2$
Best choice $w=0$.

So, $\quad y_i b \geq 1-\xi_i$
$\Rightarrow \xi_i \geq 1-y_i b \Rightarrow$ Always possible.

Note: How about $w=0, \xi_i=0$?

$\Downarrow$

$y_i b \geq +1 \ \forall i$

If $y_i=+1, b\geq +1$
If $y_i=-1, b\leq -1$ $\Big\}$ contradiction.

So the model uses $w=0, \xi_i>0$

The decision fn. is $f(x)=b \Rightarrow$ Constant classifier
- ignores the data fully.

Extreme underfitting.

Case-II. $C \to \infty$

$\min \frac{1}{2}\|w\|^2$ + huge penalty on ξ_i

So objective $\to \infty$, so it forces $\xi_i \to 0$

Constraint reduces to: $y_i(w^T x_i + b) \geq 1$

$\hookrightarrow$ Hard margin SVM!!

So, small $C \to$ flexible (ignore errors) $\to$ ignores data.
Large $C \to$ Rigid (fit all points) $\to$ enforce perfect separation.

$\xi_i = 0 \Rightarrow$ Correctly classified & on the margin
$0 < \xi_i < 1 \Rightarrow$ " " , inside the margin
$\xi_i > 1 \Rightarrow$ incorrectly classified.

Let's see the lower bounds on ξ_i

$$\left.\begin{array}{l} \xi_i \geq 0 \\ \& \ \xi_i \geq 1 - y_i(w^T x_i + b) \end{array}\right\} \Rightarrow \xi_i \geq \max\{0, 1 - y_i(w^T x_i + b)\}$$

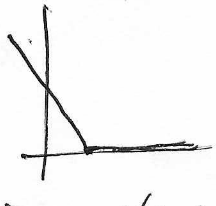
(how?)

Let, $(w^*, b^*, \xi_i^*) = \underset{w, b, \xi_i}{\arg\min} \ \frac{1}{2} w^T w + C \sum_{i=1}^{N} \xi_i$, s.t. $y_i(w^T x_i + b) \geq 1 - \xi_i$

$$\Downarrow$$

$\xi_i^* = \max\{0, 1 - y_i(w^{*T} x_i + b^*)\}$ (why?)

Hinge fn.

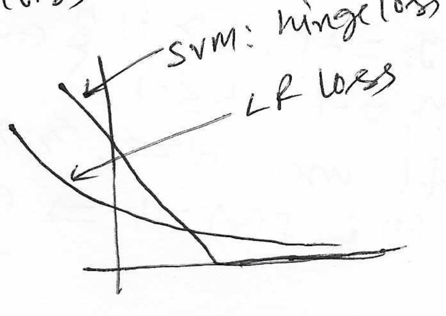


$h(z) = \max(0, 1-z)$

Now the optimization becomes:

$$\min_{\vec{w}, b} \underbrace{\frac{1}{2} w^T w}_{\text{regularization}} + C \underbrace{\sum_i \max\{0, 1 - y_i(w^T x_i + b)\}}_{\text{hinge loss } (y_i(w^T x_i + b))}$$

Logistic regression vs SVM



log loss never goes to 0.
For SVM, once you touch 0, there is no room for improvement.

SVM: $\min_w$ Hinge Loss $+ \frac{1}{2} w^T w$

LR: $\max_w \log P(y|x, w) - \lambda w^T w$